MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6AH6WA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Pentode, miniature, sharp cutoff

Outline - - - - - - - - 5-2 (EIA)
Base - - - - - - - - E7-1
Envelope - - - - - - - - T5-1/2
Cathode - - - - - - - - Coated unipotential
Base connections:

Pin No. - - -	1	2	3	4	5	6	7
Element - - -	g1	g3	h	h	a	g2	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Rk	Pp	Pg2	Ehk	TE	Rg1	Ik	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Ohms	W	W	v	°C	Meg	mAdc	ft
Maximum:	6.9	330	0, -55	165	---	---	3.5	0.5	200	+180	0.1	28	Note 1
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	300	0	150	0	160	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

Ⓖ denotes changes

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1301	Heater current			0.4	II	If	420	480	mA
1336	Heater-cathode leakage			0.4	II	Ihk	---	15	μAdc
1266	Total grid current		Rg1 = 0.1 Meg	0.4	II	Ic1	---	-1.0	μAdc
1256	Electrode current (1) (anode)	2		0.4	II	Ib	7.5	12.5	mAdc
1256	Electrode current (2) (anode)		Ec1 = -10 Vdc	0.4	II	Ib	---	30	μAdc
1256	Electrode current (screen grid)			0.4	II	Ic2	1.5	3.5	mAdc
1306	Transconductance (1)			0.4	II	Sm	7,000	11,000	μmhos
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	I	R	---	---	---
1256	Electrode current (3) (anode)		Ec1 = -5.0 Vdc	2.5	I	Ib	5	---	μAdc
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	10	%
1266	Grid emission	3	Ef = 7.5 V; Ec1 = -10 Vdc; Rg1 = 0.1 Meg (min)	2.5	I	Ic	---	-2.0	μAdc
2201	Noise and microphonics		Ef = 6.3 Vac; Ecal = 300 mVac; Rp = 10,000 ohms; Rk = 200 ohms; g1 and g3 = gnd; Rg2 = 60,000 ohms; Ecc2 = 300 V; Cg2 = 2 μF	2.5	I	---	---	---	---
1331	Direct-interelectrode capacitance		Shield No. 316 Shield No. 316 Shield No. 316	2.5	Code E	Cg1p Cin Cout	--- 8.0 3.0	0.035 12.0 6.0	pF pF pF
1031	Low-frequency vibration		Rp = 2,000 ohms	2.5	I	Ep	---	100	mVac
1041	Shock	4	450 G; Ehk = +100 Vdc	---	Note 5	---	---	---	---
1031	Vibration-fatigue			6.5	Note 5	---	---	---	---
---	Shock and vibration-fatigue-test end points:								
1031	Low-frequency vibration		Rp = 2,000 ohms	---	---	Ep	---	200	mVac
1336	Heater-cathode leakage			---	---	Ihk	---	20	μAdc
1266	Total grid current		Rg1 = 0.1 Meg	---	---	Ic1	---	-2.0	μAdc
1306	Change in transconductance (1) of individual tubes			---	---	ΔSm t	---	15	%

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 -Continued								
1121	Base strain	6		---	---	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = +100 Vdc; Ec1 = Ec2 = Ec3 = Eb = 0; 1 min "on", 4 min "off"	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	30	µAdc
1516	Stability life		Ehk = +135 Vdc; Rg1 = 0.1 Meg; TA = room	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	10	%
1501	Intermittent life	7,8	Ehk = +135 Vdc; Rg1 = 0.1 Meg; TE = +180°C (min)	---	---	---	---
---	Intermittent life-test end points (1,000 hours):						
1266	Total grid current		Rg1 = 0.1 Meg	Ic1	0	-2.0	µAdc
1301	Heater current			If	410	490	mA
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	20	%
1306	Transconductance (2)		Ef = 5.7 V	ΔSmEf	---	20	%
1211	Insulation of electrodes			R	100	---	Meg
1336	Heater-cathode leakage			Ihk	---	15	µAdc
1511	Cathode interface life	8	Ef = 6.9 V; Ehk = 0	Ri	---	25	Ohms

NOTES:

1. See "Reduced pressure (altitude) rating " and altitude, maximum peak voltage in the basic document.
2. This test shall be performed at the conclusion of the holding period.
3. Prior to this test, the tubes under test (TUT) shall be preheated a minimum of 5 minutes operating at the conditions specified below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef V	Eb Vdc	Ec1 Vdc	Ec2 Vdc	Ec3 Vdc	Rk Ohms	Rg1 Meg
7.5	300	0	150	0	160	0.1

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NOTES - Continued

4. A grid resistor of 0.1 megohm shall be added except when a thyratron-type short indicator is used.

5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

6. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (method 1121)", except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

   a. Accepted if not more than one defective for class "A", "B", or "C" defects, respectively (see method 1121), or if not more than a total of two defectives are found in the sample.

   b. Rejected if two or more defectives for class "A", "B", or "C" defects, respectively, or if a total of three or more defectives are found in the sample.

7. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurements (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

(G) 8. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

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